MAY 27 - JUNE 4, 2016

MANITOBA AEROSPACE WEEK

Strong Past, Bright Future

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FLIGHTCRAFT MAINTENANCE SERVICES REACHES NEW HEIGHTS

Thousands of Canadians every day board passenger jets destined for sunny coastal resorts or aircraft bound for friendly Arctic communities. Few of these passengers realize, however, that their aircraft could have been serviced in Winnipeg.

For more than twenty years, Winnipeg-based Flightcraft Maintenance Services has been ensuring some of Canada's most recognized airlines remain safe and reliable for customers flying to all corners of North America, and beyond.

As the only provider of heavy maintenance for large commercial aircraft in Manitoba, Flightcraft – commonly known as FMS – has close to 100 highly skilled employees and many of them have been with the company since its founding in 1995.

"FMS is evolving to meet the demands of our industry and we're expanding operations to offer new services," said company CEO Jim Peroff. "We've made significant investments in our training and our tooling which enables us to now offer all levels of aircraft maintenance including overhaul on virtually all commercial types of aircraft in North America."

Mr. Peroff notes that, with the current US dollar exchange rate, American operators are now looking north for maintenance while Canadian operators are thinking twice about going south.

Several new contracts in recent months have presented FMS with significant growth opportunities. The company recently opened line stations in Regina and Saskatoon, and the company plans to create several jobs in Winnipeg.

"FMS is pleased to have some of the best and brightest aviation techs in Canada, possibly the world, and we continue to hire talented professionals," said Mr. Peroff. "Some of these individuals were formerly working in Winnipeg's aerospace sector and left, and we're happy to see them bring their technical expertise back."

The FMS family has evolved into a dedicated and experienced team. Aviation technicians hired by FMS join a solid and growing company known for quality – meeting or surpassing all regulatory requirements in safety and airworthiness.

Their commitment to ensuring customer satisfaction and delivering results is why many clients have worked with FMS for more than two decades.

"We aim to provide world-class quality to our customers – not just in their airframes but also in our customer service," said Mr. Peroff. "Meeting our regulatory requirements is important, but we go further than simple checks in the box, and that's what sets us apart."

To manage the risks associated with growth, FMS is dedicated to detailed planning and to investing in the future. Recent investments include upgraded Technical Training, new Network Aircraft Maintenance Software and long-term access to 60,000 square feet of hangar space.

"We've recently opened a second line of narrow body Heavy Maintenance aircraft with a goal of adding a third line for wide body aircraft," said Mr. Peroff. "When you calculate the accompanying growth in support shops, our manpower requirements could easily double in the coming years."

FMS is certified by Transport Canada to perform all forms of aircraft Line and Heavy maintenance up to and including overhaul. Their business currently focuses on servicing the B737 aircraft family, which is among the world's most commonly used aircraft.

Today FMS is actively involved in surplus sales of aircraft parts where older aircraft are purchased and stripped of all useable parts. These parts are inspected, reworked, certified and made ready for resale. This is smart business, but it also allows FMS to disassemble aircraft in an environmentally responsible manner.

Any surplus part coming out of their Winnipeg warehouse is no more that 2 to 4 hours away from any North American-based customer – making their Winnipeg location a tremendous asset.

Flightcraft Maintenance Services is independently owned and operated, and has always worked out of the Winnipeg International Airport. For more information, please visit: www.flightcraftmaintenance.com ▮

A MESSAGE FROM THE CHAIRMAN

Manitoba's state-of-the-art aerospace industry is the largest in Western Canada and home to a diversified cluster of world-class manufacturing, repair and overhaul, and service firms. The aerospace industry is intensely export oriented, with 80% of $1.6 billion in sales being exported. The industry is comprised of Boeing Canada Winnipeg, StandardAero and Magellan Aerospace, as well as 23 small to medium-sized businesses. Manitoba is also home to two commercial aerospace engine test facilities, one at the Winnipeg International Airport, and the second in Thompson. The industry represents over 5,000 employees with good paying jobs.

For the last 24 years, the aerospace industry has been represented by two organizations: Manitoba Aerospace Association (MAA) and Manitoba Aerospace Human Resources Council (MAHRC). On April 1, the two organizations amalgamated and our industry is now represented by one organization, Manitoba Aerospace Inc. (MA).

The new Board of Directors of MA will build on the great tradition of the previous two organizations. The main areas of focus of each organization will merge into four areas for MA: Human Resources Development; Marketing (Branding); Innovation, Research & Development; and Development of Small to Mid-Sized Businesses (SMBs).

Manitoba Aerospace has established and maintained a significant training and education network over the past 24 years. It has helped both large and small companies access training support for company-specific training as well as multi-company training programs. It has developed strategic partnerships with post-secondary education institutions to ensure that educational programs for entry-level positions are consistent with the industry's current requirements. There are a number of career awareness programs that will continue. One of those, Aerospace & Aviation in Manitoba (AAiM Day) is profiled in this supplement.

Manitoba Aerospace, in partnership with a national organization, Consortium for Aerospace Research and Innovation in Canada (CARIC), established a regional office in 2015 and will be encouraging more research and development opportunities for companies in the industry. CARIC is also profiled in this supplement.

Kim Olson

Manitoba Aerospace will continue to develop small to mid-sized business through our Competitive Edge Supplier Development (CESD) program. In addition, we will provide support for companies to understand the federal government procurement process, specifically the Industrial Technology Benefit program.

I would like to take this opportunity to thank our departing Executive Director, Ken Webb, for his leadership, professionalism and dedication. He has done a great job representing the aerospace industry on the provincial, national and international stages. It has been a pleasure working with Ken and we wish him all the best in his "retirement."

Kim Olson, Chairman,
MA Board of Directors
Senior Vice President,
StandardAero

MANITOBA AEROSPACE

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OPPORTUNITIES AT YWG

Ian Wishart, Manitoba Minister of the Department of Education and Training

MANITOBA AEROSPACE WEEK 2016

As the aerospace industry generates quality and challenging employment opportunities for thousands of Manitobans, I am pleased that one of my first official acts as Minister of the Department of Education and Training is to proclaim Aerospace Week 2016 on behalf of the Government of Manitoba. From May 27 - June 4 Manitoba firms and their employees will promote the world class technologies, capabilities and products that enable them to compete successfully in the global marketplace. While commercial and military aircraft advanced manufacturing facilities and engine repair and overhaul activity are concentrated in Winnipeg, the provision of aerospace services occur throughout Manitoba. Thompson and Winnipeg collectively host over $100 million worth of the world's most advanced aerospace engine test infrastructure. As well, the Royal Canadian Air Force (RCAF) flight training school operates from the Southport Aerospace Centre at Portage la Prairie, in a partnership with KF Aerospace. The Manitoba Air Show will be held at Southport on June 4 and will host the RCAF Snowbirds flight demonstration team as part of Aerospace Week. In closing, I wish to congratulate Manitoba aerospace companies and the men and women who sustain their international reputation for excellence for their contribution to our community and economy. All the best with Aerospace Week 2016!

SEMAINE DE L'AÉROSPATIALE 2016

Étant donné que l'industrie aérospatiale crée des emplois stimulants et de qualité pour des milliers de Manitobaines et Manitobains, je suis heureux que l'un de mes premiers actes officiels en tant que ministre de l'Éducation et de la Formation consiste à proclamer la Semaine de l'aérospatiale en 2016 au nom du gouvernement du Manitoba. Du 27 mai au 4 juin, des entreprises du Manitoba et leur personnel feront la promotion des technologies, des capacités et des produits de niveau mondial qui leur permettent d'être hautement concurrentiels sur le marché mondial. Bien que les installations de pointe pour la fabrication d'aéronefs commerciaux et militaires ainsi que les activités de réparation et de révision de moteurs soient concentrées à Winnipeg, la prestation de services dans le secteur aérospatial a lieu un peu partout au Manitoba. Ensemble, Thompson et Winnipeg abritent une infrastructure pour les essais de moteurs aérospatiaux unique au monde et dont la valeur dépasse les 100 millions de dollars. Par ailleurs, l'école de formation au pilotage de l'Aviation royale canadienne est établie au centre aérospatial Southport, à Portage la Prairie, et a conclu un partenariat avec KF Aerospace. Le Spectacle aérien du Manitoba se déroulera le 4 juin au centre Southport. L'équipe de démonstration aérienne des Snowbirds de l'Aviation royale canadienne y participera dans le cadre de la Semaine de l'aérospatiale. Pour conclure, je souhaite féliciter les entreprises aérospatiales du Manitoba ainsi que les hommes et les femmes qui appuient leur renommée internationale. Leur contribution à notre province et à notre économie est exceptionnelle. Bonne Semaine de l'aérospatiale 2016!

EnviroTREC
Canadian Environmental Test Research & Education Center

The GLACIER/EnviroTREC facility in Thompson is a major component of Manitoba's claim to be the icing certification capital of the world for large gas turbine engines. This globally unique facility has been operational since November 2010 and has completed numerous icing certification programs.

EnviroTREC, is a not-for-profit sister organization to GLACIER. EnviroTREC's purpose is to stimulate and mentor collaborative research and development activities and to promote the development of human resources necessary to support aerospace technology development in Manitoba. A priority for EnviroTREC is the development of technologies and skills related to the test, evaluation and certification of the next generation of large aircraft engines.

Please visit the EnviroTREC website at www.envirotrec.ca for more information.

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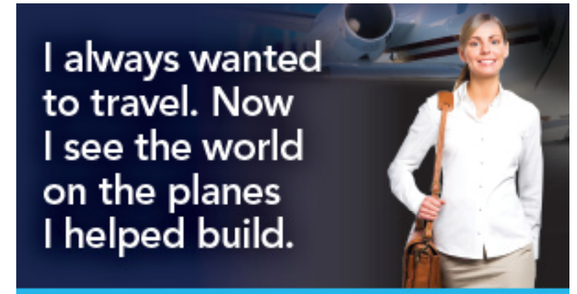

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THE SKY'S THE LIMIT

DON'T MISS THE MANITOBA AIR SHOW 2016

The Manitoba Air Show 2016 is ready for takeoff at Portage la Prairie on June 4.

While it's sure to be a fun-filled day, the Manitoba Air Show 2016 is aimed at educating and instilling a passion for flight in the next generation. Given that there has not been an air show in Manitoba of this magnitude for almost a decade, the opportunities to do so are seemingly limitless!

Led by the passion and drive of the Manitoba 99s — a chapter of the International 99s Organization of Female Pilots — and fuelled by a desire to spark a love of aviation in the hearts of our young people, our air show is sure to please. Educational tents and exhibits will be prominent features throughout the show, giving young people a chance to learn the basics of aviation, build models and see airplanes up close. They'll also have an opportunity to sign on to learn more if they like.

Heather Paddock, First Officer A320 of Air Canada, says it's important to have an air show in Manitoba, as aviation is an integral part of northern life in our province.

"The air show can be a valuable tool for young people to see first-hand what a career in aviation would involve," Paddock says. "The fact that the air show this year is being put on by the Manitoba Chapter of the 99s, a worldwide women's pilot organization, may encourage more women to pursue careers in all sectors of the aviation industry. When I was younger I always went to the Winnipeg air shows, and this may have influenced my decision to become an airline pilot."

Inspiring words from one of our finest!

This year's airshow will feature exciting aerobatics displays by the Canadian Forces Snowbirds and CF-18 Demo Team. As well, spectators will be treated to invigorating and adrenaline-pumping demonstrations by the Canadian Forces SkyHawks Parachute Team and Canadian Forces Search and Rescue Technicians, along with a CC-130 Hercules. Keep your eyes open as well for the Commonwealth Air Training Plan commemorative flypast! This year marks the 75th anniversary of the Air Cadets, and to celebrate they will be treating spectators to a glider demonstration.

In addition to these breathtaking displays of aviation and military excellence, the air show will feature a variety of static and cultural displays that are sure to create excitement for all ages. Along with displays of restored Second World War aircraft, the show will host one of Manitoba's largest-ever classic car shows. Enjoy live music, an MC College fashion show and special entertainers who will teach spectators about aviation in fun and exciting ways. See if you can spot Amelia Earhart and other aviation superstars on our tarmac!

Finally, there is a very special essay contest open to kids in Grades 4 to 12. Participants are asked to imagine what it is like to fly in an airplane and write a short, original essay, with 100 lucky winners to be selected for free airplane rides at the show! Visit www.mbairshow.com/writing-contest/ for more information on the subject!

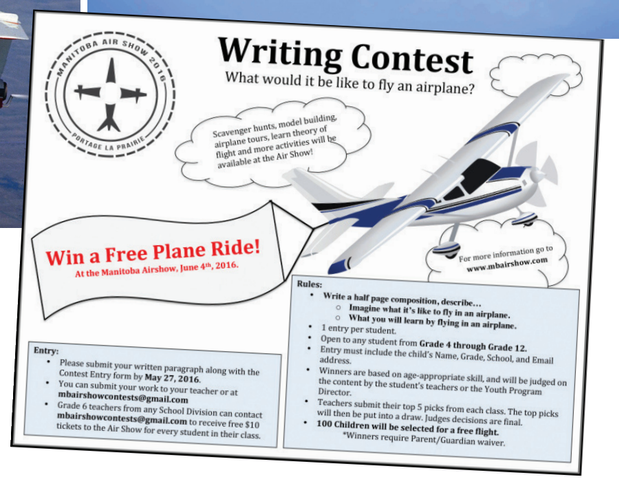
Jordan Penner, a Grade 6 teacher at Portage Collegiate, says air shows can provide a valuable experience for students.

"Canada's identity, history, culture, technology, innovation, and most certainly its future is intrinsically linked to flight. Air shows connect young people to this, which is Canada in all the best ways. At these shows our young people learn that when it comes to our future, the sky truly is the limit."

Join the 99s on June 4 as they once again set the standard in promoting women's aviation and introducing the next generation to flying!

Visit our website at www.mbairshow.com for more information and to buy your tickets! ▮

— Nadya Baspaly, Director for the Manitoba Air Show 2016.



Opposite: Western Canada 99s, including Rosella Bjornson, the first woman airline pilot in North America and founder of the Manitoba 99s, and Paulette Klassen, first woman president of the Steinbach Flying Club, with the Commonwealth Air Training Plan Museum's Harvard. The museum is offering member rides for $400-$500 at the Manitoba Air Show.

Photo by Jay Ladouceur

INSPIRING THE NEXT GENERATION

MAGELLAN AEROSPACE HELPS STUDENTS LAUNCH SCIENCE INTO SPACE

Magellan Aerospace believes strongly in the importance of youth and learning. We recognize the importance of education; especially science, technology, engineering and mathematics (STEM) education, which is the groundwork for the next generation of technologists, engineers, and scientists.

Magellan has a long demonstrated history with supporting academia and student programs, some of which include Aviation and Aerospace in Manitoba (AAiM) Day, Make Your Move, which is presented by WISE Kid-netic Energy, and the University of Manitoba's SAE program. Magellan's Winnipeg plant is also home to two collaborative classrooms — the Advanced Satellite Integration Facility (ASIF) with the University of Manitoba, and the Centre for Non-Destructive Inspection (CNDI) with Red River College.

The most recent program that Magellan has thrown our support behind is called the Student Spaceflight Experiments Program (SSEP). The SSEP is an immersive educational program that provides students the opportunity to design and propose real microgravity experiments that will be flown into low earth orbit on the International Space Station (ISS). The program was created to promote STEM learning in schools and Magellan is very pleased to be able to contribute in a meaningful way to bringing the SSEP experience to Canadian youth.

"Magellan Aerospace is proud to be a part of such an exciting program," says Paul Heide, General Manager, Magellan Aerospace, Winnipeg.

"This program brings real-life space experimentation right into the classroom, getting the next generation excited about engineering, space science and STEM learning."

This unique, real-life, hands-on program was launched in June 2010 by the National Center for Earth and Space Science Education (NCESSE). It provides (typically) 300-plus students per community the chance to engage in every facet of developing real science experiments for space and aims to change the way students perceive science and their ability to do it. After several extremely successful missions in the U.S., the SSEP was extended to Canada in 2012 via the NCESSE's new international arm through the Arthur C. Clarke Institute for Space Education.

When the SSEP was broadened to include Canadian participation, two school districts (or communities of schools) were able to participate in this unique program and send scientific experiments into space — Stonewall, Manitoba on Mission 3 to the ISS in 2013, and Kamloops/Thompson School District 73, British Columbia on Mission 6 to ISS in 2014.

In order to participate in the program, the communities were required to fundraise in order to cover some of the costs of the space launch. Magellan participated on the funding teams for both of these Canadian student missions, and has experienced first-hand the excitement that this once-in-a-lifetime learning opportunity delivers to students.

"The transformation in the students in the Interlake has been remarkable as well as the students in Manitoba as a whole. SSEP is a truly remarkable program that encompasses all of those skills and more," says Maria Nickel, SSEP Director - Interlake, Stonewall, Manitoba.

"It has done for kids science in Manitoba what Chris Hadfield did for the world: (inspire) wonder about our universe and curiosity about science."

In becoming an international partner for the program in the spring of 2015, Magellan increases the opportunity for more communities to participate in the SSEP. The program uses the partner funding provided by Magellan to bridge funding shortfalls for communities that may otherwise be unable to participate. Magellan sees this funding as an investment in Canadian youth, and in nurturing an interest in science to the next generation of scientists and engineers.

On the success of the first two Canadian programs, the SSEP has taken root, gaining more and more visibility and interest throughout the country. Early in 2015, three more Canadian communities were selected for participation in the program and the opportunity to see their science fly to space on SSEP Mission 8. The Magellan funding commitment has helped two of these Canadian communities — Ryerson University and Local High Schools in the Toronto District School Board (TDSB); and the University of Toronto Schools (UTS).

These two communities immersed 375 students in microgravity experiment design, resulting in 56 flight experiment proposals submitted.

"When my team heard that we won, we were in disbelief," says Francis Buguis, a member of the TDSB team. "It is hard to believe that something I have worked on in my second year of university will actually reach space. This has been a fantastic experience, I have learned a lot in the time I spent researching for our experiment. It will be really exciting to continue with this endeavour."

Magellan's partnership funding was also required for two of the Mission 9 communities to go forward and participate in the program — Langevin School, North Calgary, Alberta, and West Vancouver School District, Vancouver, British Columbia. These two communities are expected to engage about 850 students in microgravity experiment design.

The four current (Mission 8 and 9) Magellan-sponsored student groups are on the countdown to launch right now.

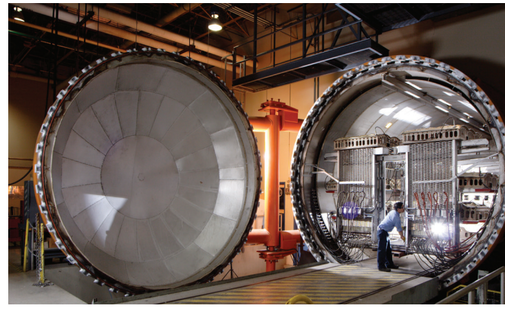
As future missions get underway, the support provided by Magellan will continue to be crucial for enabling a greater number of students access to the SSEP program; exposing them to ideas and possibilities they may not have been aware of, and keeping them engaged and active in STEM learning.

The sky is the limit for this next generation of space scientists and explorers! ▮

Above: The Interlake School Division team examines their experiment after receiving it back from the International Space Station.

Photo courtesy of Magellan Aerospace, Winnipeg

BOEING AND EMPLOYEES CONTINUE TO SOAR



In its nearly 100 years of operation, The Boeing Company has continued to provide commercial airlines with some of the latest advances in aerospace technology. Those innovations have included everything from the composite materials used in the wings and fuselage of its Boeing 787 Dreamliner to a new hybrid-electric propulsion system currently being tested.

One thing that hasn't changed during that time is the company's focus on supporting its employees, including those employed at its Winnipeg site headquarters in Murray Industrial Park. Just ask Shirley Langver, who has been with the company 38 years and worked her way up from an entry-level position to become a senior manager. Langver was first diagnosed with cancer back in 2005 and battled the disease on four separate occasions. Today she is cancer free and credits the support of her employer and co-workers with helping her win her fight.

"I can honestly tell you the support of (people) here and the strength I pulled from that… was a big part of what pulled me through my struggles," she said. "That is very important to me, the relationships I've built over the years and the friendships I have here at Boeing."

That kind of support is a big reason why Winnipeg-based Boeing Canada Operations Ltd. has regularly been ranked among Manitoba's Top 25 Employers.

Since it opened for business in 1971, Boeing Winnipeg has grown from 57 employees to more than 1,600, making it one of the city's largest employers. Perhaps an even more impressive number is the average length of service for employees at the company is 20 years.

Terry Trupp, a communications specialist who has been with Boeing for 28 years, says one of the main reasons employees tend to remain with the company is its efforts to actively engage them in the decision-making process.

Employee teams have a say on everything from how to make their work environment safer to how to reduce waste. Last year, a team working on the Boeing 787 Dreamliner recommended switching from using disposable batteries in their headlamps to rechargeables. The switch resulted in a reduction of the number of batteries the team uses from 50,000 to just 400 per year and reduced battery waste from 18,000 pounds a year to about 2,000.

"Rather than having it top down, they're empowered and they empower us to make improvements. That's really helped keep us strong and moving forward," Trupp said.

There is very much a family feel among staff at Boeing Winnipeg, both literally and figuratively. In fact, it's not uncommon for multiple generations of the same family to work together.

Trupp said the company has made a concerted effort to create a family-like atmosphere where people know and trust each other. It provides a fully-furnished gym that is available to staff for a nominal fee, organizes employee baseball and curling leagues and arranges annual Christmas parties for staff and their children.

The company has also made a commitment to investing in the long-term development of employees. It offers tuition subsidies for certificates as well as undergraduate and graduate degree programs and financial assistance for individuals working towards their professional accreditation. In addition, Boeing offers a two-year Leadership Development Excellence (LDE) program designed to develop future leaders within its ranks and in the community. The company also provides local university and college students with on-the-job experience through paid internships which includes mentorship for all students.

Trupp said one of the things she is proudest of is the company's efforts to celebrate diversity in the workplace. Each year it stages Diversity Week, a mini Folklorama-like celebration during which employees get to sample all kinds of ethnic food and culture. The company also provides a variety of communications tools to deaf employees and ensures there are interpreters at all meetings they attend.

There are also a few days each year set aside to celebrate Aboriginal culture with an event known as Aboriginal Days involving cultural displays, entertainment and food.

Even as its Chicago-based parent company prepares to celebrate its past during this year's centennial anniversary celebrations, Trupp said Boeing employees continue to look ahead to Boeing's second century. That future will include more innovation while maintaining current priorities like the many charitable efforts Boeing employees are involved in such as serving breakfast at Siloam Mission, working with students at Fort Whyte farm and the annual United Way plane pull and employee charity campaign. ▮

Boeing Canada Winnipeg employees at Canada's largest composite manufacturing site.

TRAINING DAYS

StandardAero gives students a head start

By Amanda Schultz

StandardAero partners with Tec Voc to provide students with Initial Human Factors training.

Photo courtesy of StandardAero

There was a typical Manitoba blizzard on the evening of Dec. 10, 2013. Insignificant to most, but one that I will never forget. The following morning, I was to teach 50 students enrolled in both the Aerospace Technologies and the Aerospace Manufacturing and Maintenance Orientation Program (AMMOP) at Tec Voc High School.

In an effort to broaden the employability of students graduating from these programs, StandardAero and Tec Voc established a partnership to provide them with their Initial Human Factors training. My time as a certified instructor was donated to teach on behalf of the Canadian Council for Aviation and Aerospace (CCAA).

Transport Canada requires all technical employees within the aviation and aerospace industry to complete Initial Human Factors training within their first year of employment. This course covers topics such as Human Performance, Error Management and the Dirty Dozen, or the 12 most common factors that lead to mistakes within the workplace. This certification must be renewed every three years thereafter. By providing these students with their initial training prior to entering the workforce, we reduce the cost to future employers.

Back to the blizzard … After unloading the massive amount of course materials into a snowbank and carrying them up two flights of stairs, I managed to locate my classroom.

Normally, I teach this course to a class size of 16-20 employees at StandardAero. At Tec Voc, I walked into a classroom of 50 students. Slightly overwhelmed, I did my best to make the classroom activities work for such a large group. I was very impressed with the level of professionalism of the students. They were very eager to learn and listened attentively to stories about my experiences working within the aviation industry.

I noticed that the students were able to relate the material back to all industries, rather than view it solely as an aviation industry requirement, and find the value in it. Many of these students have experience working in the hospitality, retail or fast food industry. Through discussion, we were able to link the causes of various accidents back to the Dirty Dozen, regardless of the industry. Feedback from the students actually recommended that the material be taught in all workplaces.

Teaching these students has been an extremely rewarding experience for me. December 2015 marked my third year of instructing with Tec Voc and I look forward to all future opportunities to do so — although I think the biggest reward is recognizing a new hire on the shop floor who was part of this program. ▮

— Amanda Schultz is Senior Learning & Development Specialist at StandardAero

THE DIRTY DOZEN

The 12 most common factors that lead to mistakes in the workplace:

Lack of Teamwork
Lack of Communication
Lack of Assertiveness
Fatigue
Stress
Distraction
Lack of Awareness
Lack of Knowledge
Lack of Resources
Pressure
Negative Norms
Complacency

Where custom solutions come Standard.

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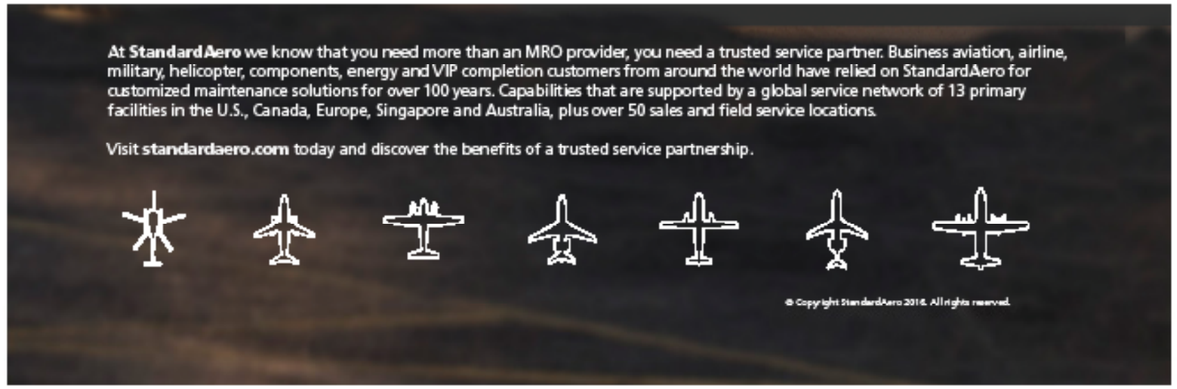


Girls and their parents taking part in Girls in Aviation Day activities at the 2015 event. Women in Aviation International chapters around the world present Girls in Aviation Day on Sept. 24.

GIRLS IN AVIATION DAY

Join us on Saturday, Sept. 24, 2016 at the Lyncrest Airport, where Northern Spirit, the Winnipeg Chapter of Women in Aviation International (WAI), will be hosting their second annual Girls in Aviation Day.

This is an exciting event where girls can spend time with women who are leaders and role models in the aviation and aerospace industries right here in our community. On that same day, Women in Aviation International chapters are hosting unique events around the world and Northern Spirit is proud to be a part of this inspirational program.

Winnipeg has a long and successful history in aviation and aerospace, and there are many rewarding career opportunities and educational pathways to those careers. Despite all of these opportunities, the aviation and aerospace industries are largely dominated by men: Only 15% of air traffic controllers, 5.4% of airline pilots and a mere 2.5% of aircraft maintenance engineers are women.

"There are no bounds to which girls can pursue and excel in these rewarding fields, and we want girls to know that these careers are available to them as well," says Kristin Long, the Northern Spirit Chapter President, and 737 Captain.

In an effort to introduce girls to the exciting opportunities available to them, Women in Aviation International has designed a fun and interactive event for girls aged 8-16. The event will include interactive activities at seven different career stations, and the girls will have the opportunity to meet a pilot, flight attendant, air traffic controller, airline dispatcher, maintenance engineer, airport operations manager and aerospace engineer. At every station, the girls will engage in a hands-on activity and learn about each specific career.

"It's all about having fun and discovering the cool and exciting world of aviation and aerospace," Long says.

This year, the girls will be able to get outside on the airfield at the Lyncrest Airport and really discover the world of aviation and aerospace.

There will be aircraft on the ramp that participants will be able to touch, climb in and explore. In addition, there will be inspirational speakers and a Career Exhibition where local organizations, training programs, and educational institutions will be on hand to help the girls navigate their potential future careers in aviation and aerospace.

Women in Aviation International is proud to offer a new WAI Aviation Girl fun patch for Girl Guides who attend Girls in Aviation Day events around the world. All Girl Guides who attend our Winnipeg event will earn this new badge that they can wear proudly. ▮

Registration is now open, and you can register directly from our website at www.northernspiritywg.ca/girls-in-aviation-day.html.

More than 80% of Royal Canadian Air Force (RCAF) pilots receive training at the Southport facility operated by KF Aerospace.

Photos courtesy of KF Aerospace

TRAINING GROUND

KF AEROSPACE PUTS PILOTS ON CAREER FLIGHT PATH

By Pat Rediger

Many people are surprised to learn that 80% of pilots in the Royal Canadian Air Force (RCAF) get their wings in Portage la Prairie.

The Contracted Flying Training and Support Training Centre at Southport Airport is operated by KF Aerospace.

It's here that primary pilot training and selection takes place and it's also where advanced training is provided for pilots destined to fly helicopters and multi-engine, fixed-wing aircrafts.

"I think people tend to be more familiar with the Air Force base in Moose Jaw – being that it's the home of the Snowbirds – and think of it as the big training base in Canada. It certainly is, but as far as output for the Canadian Forces, a lot happens here at Portage," says Bill Ryan, marketing manager for KF Aerospace. "We're a hidden secret."

On Oct. 27, 2005, KF Aerospace was awarded a 22-year contract by the Department of National Defence to provide training support to the RCAF. KF Aerospace leads a joint partnership that includes Canadian Helicopters, Bluedrop Performance Learning and Canadian Base Operators.

The history of aviation in Portage, however, goes back all the way to the Second World War, when the Royal Canadian Air Force Station served as home of the No. 14 Elementary Flying Training School, working with air crew trainees from countries including Canada, France and Czechoslovakia (today, the main building of the training centre is named after Mark Henry Brown, who was the first Canadian flying ace of the Second World War). Closed shortly after the war, the station was reopened as part of a rapidly expanding program to train NATO pilots during the Cold War.

"Today's state-of-the-art facilities, training devices and operations reflect a long and proud tradition of excellence in aviation training dating back to WWII and the British Commonwealth Air Training Plan," says Ryan.

KF Aerospace operates four fleets of aircraft: The Grob G120A for Primary and Basic training, the C-90B King Air for Advanced Multi-Engine training, and the B206 Jet Ranger and B412 CF Outlaw to support Advanced Rotary Wing training. KF Aerospace and its partners provide all the ground school and simulator instruction, flight instruction on the Grob G120A, and all of the support services for the program.

Ryan says it all begins with Phase I Primary training, which represents a critically important element of the RCAF Pilot Training System. Pilot candidates receive their introduction to RCAF flying training and they fly their first training missions. As well as assessing basic pilot skills, the selection process is further refined with the aim of identifying those candidates who are likely to achieve the required pilot training standards within the allotted training time.

"Early identification of potential to succeed is necessary to provide the candidates with the confidence to achieve success in future pilot training phases, and also to ensure the substantial investment required for Phase II Basic and Phase III Advanced flight training are based on a solid foundation," Ryan says.

Upon completion of Phase I Primary training, most successful students proceed to 2 Canadian Forces Flying Training School in Moose Jaw, Sask., for Phase II training on the Beechcraft T-6A Harvard II aircraft. At the end of the Phase II program, students are streamed to Advanced Training on Jet, Rotary Wing or Multi-Engine Fixed Wing aircraft. All those selected for the rotary wing or multi-engine programs return to Southport.

KF Aerospace takes full advantage of the latest advances in simulators and course equipment to enhance their students' experience.

"We've come a long way from the days of overhead projectors, chalkboards and slide projectors," Ryan says. "Our classrooms are equipped with the latest in learning technologies such as SMART Boards that take full advantage of modern computer-based modules and multi-media options. Ground-based training includes a wide range of computer-based instructional modules, available online to students on their laptops 24/7," he adds.

"The next step from classroom instruction to flying the actual aircraft takes students to Flight Training Devices, where they can practice normal and emergency operations on their specific aircraft."

Ryan notes that a perfect example of this technology can be found in the Phase III Multi-Engine program.

"An extremely important element during this phase of training is the state-of-the-art Level D full-motion simulator, with high-end graphics," he says. "This simulator provides a full range of training scenarios, including advanced visual flight rules conditions, a full complement of instrument flight rules missions, and in-depth training in correctly handling both critical and non-critical ground and in-flight emergency situations, some of which could not safely be taught on the aircraft."

More than 100 pilots are in training at any given time, supported by approximately 250 employees who do everything from direct training to air traffic control to providing meal service. ▮

AAiM DAY CELEBRATES ITS 10TH ANNIVERSARY

Friday, May 27, 2016, will mark a milestone with the 10th annual Aerospace & Aviation in Manitoba (AAiM) Day program, which brings together the local aerospace and aviation communities to provide a learning experience for students in Grade 6.

The one-day event at Red River College - Stevenson Campus is designed to provide career awareness activities that relate to the aviation and aerospace industries and correlate with the Manitoba Grade 6 science curriculum units on flight and space science. Since its inception in 2007, the program has grown to include up to 700 students from both urban and rural schools as well as private and home-school organizations.

More than 150 volunteers from the various host organizations make the event possible.

AAIM DAY HOST ORGANIZATIONS ARE:

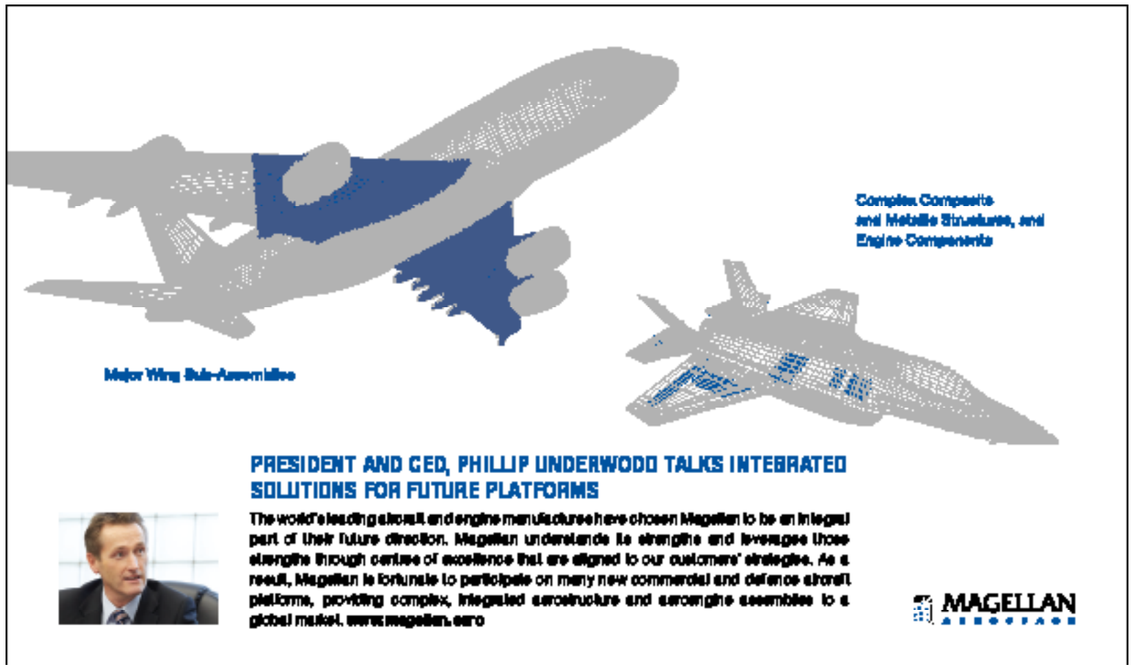
- Red River College - Stevenson Campus
- Boeing Canada Winnipeg
- Magellan Aerospace
- StandardAero
- Manitoba Aerospace
- Career Trek
- Manitoba Rocketry Group
- Tec Voc High School
- Women in Science and Engineering (WISE), University of Manitoba
- 17 Wing (Canadian Air Force)
- Canadian Forces School of Aerospace Studies
- Air Cadet League of Canada (MB) Inc.

The program runs in two sections, with approximately 350 students participating in six, 15-minute hands-on activities in both the morning and afternoon sessions.

PROGRAM ACTIVITIES ARE:

- Aircraft Inspection: Students learn to identify parts of aircraft by taking on the roles of aircraft maintenance engineers on inspection
- Characteristics of Flight: Students learn about the characteristics of flight, then design and fly a small paper glider
- Composite Materials: Students learn about the latest composite materials used in aircraft and create a composite part
- Hercules Aircraft Theatre: Students learn about the various aviation and aerospace careers by taking part in an imaginary search and rescue flight
- Rockets: Students learn about rockets in order to design and fly a small paper rocket
- Space Crater Fun: Students learn about craters and space science by creating a variety of simulated craters

AAiM Day is an exciting and fun-filled educational experience for all the participants, and a chance for the host organizations to make a significant impact on a large number of young people who, over the years, have learned a bit more about opportunities in aerospace and aviation in Winnipeg. KUDOS TO ALL INVOLVED!



COPA FOR KIDS FUELS DREAMS OF FLIGHT

During Manitoba Aerospace Week, there is one important factor that should not be overlooked. The aviation industry needs well-trained and well-equipped personnel to do the job, and it needs well-trained and well-motivated pilots.

Often, becoming a pilot is something people dream of at a very young age. The 17,000-member Canadian Owners and Pilots Association (COPA) developed a program that captures a young person's desire to soar like an eagle, but in modern, sophisticated and well-equipped airplanes.

To complement the successful Air Cadet glider and power aircraft program, COPA administers its own program, called COPA For Kids. The mission for the COPA For Kids Aviation Program is to provide a motivational aviation experience, focusing on a demonstration flight in an airplane.

Flights are provided free of charge to any youth aged 8-17 who wishes to participate. Since its inception in 2009 more than 10,000 young aviators have experienced the thrill of first flight.

Many of COPA's National Flights (local chapters) participate in COPA For Kids, and for many, flying young people has become a highlight of the summer season. Pilots volunteer their airplanes, their fuel and their time, while other members organize events for hundreds of young people to come out on a given day. One measure of the success of the program is that many people who had an introduction to flying at an early age have gone on to get their pilot licences and are now flying young people themselves.

The COPA For Kids program is sponsored by the insurance underwriter AIG. It was created as a Canada-only program for COPA Flights and is similar to, and has been inspired by, an equivalent program in the United States and around the world called Young Eagles, which was created in 1992 by the Experimental Aircraft Association (EAA) Aviation Foundation. COPA is not associated with the EAA Young Eagles Program. Both EAA and COPA applaud one another's efforts to involve young people in aviation, and each organization offers its members their own way to demonstrate the wonders of flight to young people.

COPA pilots retain a record of COPA For Kids flights and register the flight in the COPA Junior Aviator Logbook. Each young person who participates will receive a pilot's log book in which their first flight is entered and can check the Junior Aviator Logbook on the COPA For Kids website to look for his/her name. To learn more, visit www.copaforkids.org. ∎

— Jerry Roehr, COPA Treasurer and Director, representing Manitoba and Nunavut.

JOIN THE FUN AT RCAF RUN 2016

RCAF – 17 WING WINNIPEG WELCOMES THE ENTIRE COMMUNITY

If you have ever wondered about those big grey aircraft that fly over the city on a regular basis, you now have an opportunity to get up close, talk to some of the crew members and discover 17 Wing Winnipeg.

On Sunday, June 5, the annual RCAF Run features timed distances, a 3km Family Fun Run/Walk and lot of additional activities and entertainment, says RCAF Run Chair Lt.-Col. Danielle Clouter.

"In addition to interactive aircraft displays along the flight line, including the full-size interactive CF-18 cockpit, there will be children's activities, an interesting Race Village, athletes from local professional sports teams and much more, including two special features this year," says Lt.-Col. Clouter.

"The Snowbirds Demonstration Team will highlight the race start with a flypast and there will be one or two of the distinctive red and white aircraft on display, along with a CF-18 fighter jet with pilot."

Both will be in Manitoba as part of the Manitoba Air Show in Portage la Prairie on June 4 and are staying over to take part in the RCAF Run festivities.

The RCAF Run is for everyone — no matter what their fitness level or ability to actually run. It is a part of the Canadian Armed Forces commitment to fitness and healthy living and anyone who wants to share in that commitment is welcome to hit the tarmac with the men and women who daily serve and protect Canadians from coast to coast to coast.

The runners in timed events will be off first and these aspects of the race are certified by Athletics Canada and accredited by the Manitoba Runner's Association. The RCAF Run has become an attractive race option for serious runners who like the course and the race organization.

Anyone who wishes to take part in any aspect of the race who requires a service dog or support runner is most welcome, says Lt.-Col. Clouter.

"This is a run that is truly for anyone who wishes to take part."

For families, or indeed for anyone wishing to take part in the 3km Family Fun Run/Walk, this event offers a great opportunity. Not only can families stroll along the flight line where a selection of aircraft will be parked; they will also be able to get up close to other planes, take a tour and chat with the men and women who fly in them.

At the end of the run or walk, everyone is encouraged to take time to visit the Race Village, which features activities for all ages. A light lunch is available to participants and there will be a spirited band concert by Jet Stream from the RCAF Band.

If you are interested in taking part in race day, check out the website at www.rcafrun.ca. ∎

GOING ABOVE AND BEYOND

BOEING CANADA WINNIPEG RECOGNIZES CORMER GROUP INDUSTRIES FOR GREAT WORK

By Pat Rediger

It may have taken some late nights, and a few weekends were sacrificed, but Cormer Group Industries got the job done.

In early 2015, the Winnipeg multi-national manufacturer of world-class machine details and assemblies completed a 787 Dreamliner airplane retrofit project for their client, Boeing Canada Winnipeg, in a remarkably fast and efficient manner.

On April 25, 2016, Boeing visited Cormer and held a special ceremony to show its appreciation.

"We were very surprised to be recognized," says Leo Sousa, President of Cormer Group Industries. "Boeing presented us with a certificate and a model of the 787 airplane to put in our display stand. Our employees were proudly taking pictures."

Boeing is the world's largest aerospace company. Boeing Canada Winnipeg was looking for a company to support short-flow requirements for various new components for the 787 Dreamliner — a long-range, mid-size wide-body, twin-engine jet airliner — in order to meet its manufacturing commitments to Boeing South Carolina for the 787 program.

When the contract was awarded to Cormer in January of 2015, speed was of the essence.

"They came to us saying, 'We're on a deadline and you need to rise to the challenge,' " Sousa says. "We had to add additional resources and manpower in a very condensed timeline. That was really the biggest challenge. It was also a matter of reprioritizing all of our customer commitments and ensuring that our current customers and current work orders didn't fall behind schedule."

Cormer worked hard to dramatically improve the first article part delivery and also significantly improved the first production shipment. Sousa notes that the project should have taken around five months, but Cormer completed the work in about three.

"Two months doesn't sound like an awful lot of time, but the fact that we were able to bring it all together on a tight deadline was pretty impressive," he says. "We completed the project right the first time, which is pretty rare when you're tackling a project of this magnitude."

Sousa adds that the tight deadline also put a strain on the company's outside suppliers, but they stepped up to the plate.

"This was a great demonstration of collaboration among local companies in the industry," he says. "Without the local collaboration, being able to coordinate and visit each other with challenges, or talk to each other quite readily, this project probably wouldn't have been as quick and successful as it was."

For Cormer, this project was part of an ongoing relationship with Boeing.

"We work closely with Boeing and we have regular meetings," Sousa says.

"We had a relationship with them and had proven ourselves over the course of many years, which is why they turned to us in a pinch. It had been some time since we have launched a new product for them, so everybody felt good about it." ▮

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SALUTING THE ALL-STARS

The recipients of the 14th annual Manitoba Aerospace All-Stars Awards of Excellence were celebrated for excellence in education and training; leadership; innovative businesses practices and teamwork — and for helping to build the aerospace community at large.

• The Builder Award, which recognizes contributions made to the benefit of the entire membership of the organization and the interests of the provincial aerospace industry, was awarded to Rick Jensen, who retired in 2015 after a 43-year career with Boeing Canada Winnipeg.

Serving in multiple roles at all levels, Rick helped Boeing Winnipeg develop from a small plant of 57 skilled workers in 1971 to Canada's largest composite components manufacturing facility, employing 1,600 people in 2015. He was instrumental in advancing the scope and complexity of Boeing Winnipeg's manufacturing capability and influential in gaining design authority for the 787 Dreamliner.

Inside and outside of Boeing, Rick worked to help grow the Manitoba aerospace community. He was a champion of the Composites Innovation Centre (CIC) and served on the boards of the CIC, the Manitoba Aerospace Association (MAA), and the Aerospace Industries Association of Canada (AIAC). He led Boeing Winnipeg's community relations initiatives, helping Boeing to build stronger communities and a better world. Under his leadership Boeing become a consistent Manitoba Top 25 and a Canada Top Diversity Employer, and the first-ever recipient of the CME Community Service Award.

• Leo Sousa received a Leadership Award for leading Cormer Group Industries' expansion into an international aerospace and defence enterprise, and for helping to grow Manitoba as a hub of aerospace and defence excellence.

Joining Cormer in 2008 as Senior Finance Manager, Leo led the development of corporate strategies, business transformation, and market access. Since becoming President of the Cormer Group of Companies, he has helped diversify and increase the number of customers, opened new plants in New Brunswick and Mexico, and brought new international customers to Manitoba. Under his leadership Cormer has been recognized as one of Canada's top-ranked defence companies for the last four years by Canada Defence Review.

Leo ensures that he spends time on the shop floor meeting and listening to employees, and in the community, supporting community-based and not-for-profit organizations.

• Barb Bowen, Special Programs Manager with Manitoba Aerospace, received an award for excellence in Education and Training.

Barb joined Manitoba Aerospace — formerly Manitoba Aerospace Human Resources Council (MAHRC) — in 1999 and since that time has amassed a truly amazing body of work with educators, educational institutions, student groups, and organizations. Barb was instrumental in providing people with opportunities to develop the required knowledge and skills to gain employment in the aerospace industry, and in reaching out to students and youth to inspire them to pursue a career in aerospace.

Barb has either led or contributed her efforts to a number of highly successful outreach programs, including: Career Trek, Women in Science and Engineering (WISE), Manitoba Space Adventure Camp, WinCube, Aviation and Aerospace in Manitoba (AAiM) Day, Girls in Aviation Day, and a host of career fairs. Her outreach to women, the Aboriginal community, and youth is essential in bringing diversity to the workforce, now and for the future.

Top: StandardAero eShop Order Implementation Team. Left to Right: Barb Bowen, Manitoba Aerospace Human Resources Council (MAHRC); Leo Sousa, Cormer Group Industries; Rick Jensen, Boeing Canada Winnipeg (retired).

• StandardAero's eShop Order Implementation Team was recognized for excellence in Teamwork and Innovative Business Practices.

The StandardAero eShop Order project involved converting the internal paper-based shop order system to a fully electronic shop routing system, improving quality, reducing cycle and turnaround time, and reducing costs.

The amount of paper documents was reduced by 80 to 90%, significantly reducing paper and printing requirements as well as eliminating the costs to file, sort, store, maintain, ship, retrieve and track physical records.

The most valuable improvement, however, is the improvement in quality control in a system that accepts only appropriately trained and authorized employees to perform and sign-off tasks. This alone reduced the time required to complete the final inspection process by nearly 50%.

This complex project required the participation of people across all levels of the organization. The 20-member interdisciplinary team turned the vision into reality through collaborative teamwork and relentless project management.

The team included Tyler Ross, Jennie Grobb, Peter Awender, Brett McCullough, April Scrapneck, Ray Janzen, Chris Russell, Dana Hartley, Leroy Lemon, Mike Lepage, Gerry Lesnick, Peter Wheatley, Dave Smith, Derek Ashcroft, Amanda Schultz, Tom Scheller, Corwin Collins, Bryan Riley and Bev Picton. ▮

The Manitoba Aerospace All-Stars Dinner and Awards evening is also an important fundraiser for the Aerospace Student Awards Endowment fund. Each year, the endowments support six financial awards to high school and post-secondary students, helping them access rewarding, enduring careers in the aviation and aerospace industries.

West Canitest R&D Inc. (WestCaRD) engine test facility at the Winnipeg International Airport.

FUTURE SENSE

THE MANITOBA AEROSPACE SENSOR CENTRE OF EXCELLENCE

Humans rely on their five senses for every aspect of day-to-day life. The data streams for these five senses are supplied by billions of cellular-size sensors providing information that is sourced and processed in real time. While physical systems can't match human capabilities, advanced materials and processors are enabling the specialization and miniaturization of advanced sensors that provide massive amounts of data to faster and more complex processors.

Sensors are increasingly at the core of our ever-more automated world. Cars that autonomously navigate across the country and factories that operate 24-hour shifts all rely on advanced sensors for guidance.

Advanced sensor application and development is at the core of new business activities taking place in Manitoba. Few people are aware that approximately 85% of the world's new large commercial aircraft developmental engines are operated in Manitoba test facilities before they are approved for flight. Test and certification facilities in Thompson and in Winnipeg form a critical link in the development and certification process for General Electric, Pratt & Whitney, and Rolls-Royce aircraft engines, the world's three largest aircraft engine manufacturers. These firms have invested $100 million in these two facilities in order to perform various engineering tests, including icing ingestion.

During developmental and certification testing, the accurate measurement and processing of a large number of engine parameters, often sampled thousands of times a second, is foundational to ensuring flight safety, operational economies and reducing the environmental impact of these extremely complex systems. The globally unique facilities in Winnipeg and Thompson stand out as truly impressive engineering feats. However, the advanced sensor technologies at their core are key to realizing new opportunities both inside aerospace and in the many other industry sectors driving Manitoba's robust, diverse and growing economy. One of those opportunities lies in sensor research, development and manufacturing in Manitoba.

Building from the icing research and development experience, West Canitest R&D Inc. (WestCaRD) initiated the Manitoba Aerospace Sensor Centre of Excellence in September 2015 to build a joint industry-government-academic framework focusing on advanced sensing technologies. The founding parties include the National Research Council of Canada, the University of Manitoba, GE Aviation and WestCaRD. This activity builds on GE Aviation's strong and cooperative presence in Winnipeg. While a key portion of this initiative will continue to focus on GE Aviation's needs for engineering, development, and safety of flight certification activities, it has also identified an opportunity for a broader multi-sector initiative targeted at sensors for manufacturing needs beyond those of just aerospace.

While the initial steps were focused on aerospace, and specifically aero-engine data requirements, the next steps will explore the sensor and sensor integration technologies for the "factories of the future" or "Industry 4.0." Industry 4.0 will see humans and autonomous robots working safely and collaboratively, seamlessly designing, selecting or machining sub-components, assembling, and then shipping products around the world. To reach its ultimate potential this process must be fully integrated and must be able to correct errors and inefficiencies on the fly. These future products, such as engines, will employ sensors during manufacture and then use those same sensors to monitor their operation in service. System health during operations will be communicated to the manufacturer, who will likely draw more revenues in servicing those products than in selling them.

Every time you take off in a new airliner, you are counting on the excellent work done, in part in Manitoba, to ensure that you safely reach your destination. Now, the Manitoba Aerospace Sensor Centre of Excellence partners, supported by effective and far-sighted government involvement, are exploiting the province's niche industrial and academic strengths to develop sensors for the factories of the future right here in Manitoba.

Stay tuned as advanced sensor technology workshops are planned for 2016 in Ottawa and Winnipeg that will further explore and expand this exciting initiative. ▮

— Bob Hastings CD P.Eng., CEO West Canitest R&D Inc.

ENABLING AEROSPACE INNOVATION IN MANITOBA

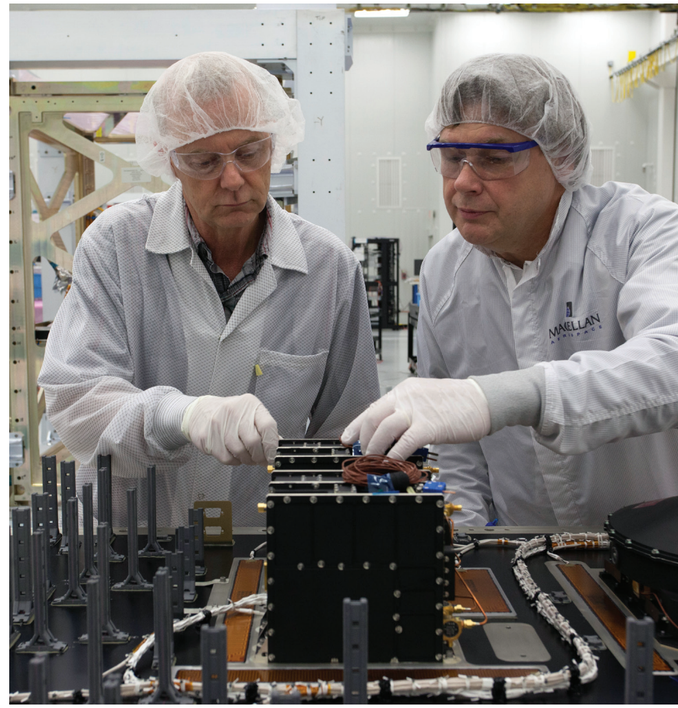
Historically, aerospace in Canada has "punched above its weight" in terms of market share vs our population. This is especially true in Manitoba, where our high-technology products and services compete directly in a competitive global market. A detailed study of the competitiveness of the Canadian aerospace industry (the Emerson report) highlighted the benefits of collaborative research and development (R&D), since aerospace is much more R&D-intensive than most other industries.

From the Emerson study recommendations, the Consortium of Aerospace Research and Innovation in Canada (CARIC) was created to provide a national network for R&D collaboration and support, including funding. Locally, Manitoba Aerospace has championed an Aerospace Technology Roadmap and worked to increase the amount of aerospace R&D in Manitoba. The collaboration of these two organizations has injected new funding for R&D projects and is just getting started.

Manitoba Aerospace hosts the central region CARIC office and in its first year of operation CARIC has already supported two major research projects, and a number of sponsorships and awards at the University of Manitoba. The first CARIC project to be approved from Western Canada is a $2.5-million collaborative project coordinated through the Canadian Composite Manufacturing R&D Consortium (CCMRD), a consortium of aerospace companies and research organizations across Canada.

Six industrial partners, including Magellan Aerospace and Boeing Canada Winnipeg; two research organizations, including the Composites Innovation Centre (CIC); and two academic partners, including Red River College (RRC), will collaborate to develop innovative solutions in design, technology development and manufacturing processes to produce challenging aircraft structures from composite materials for future applications.

In January, the StandardAero-led research project "Evaluation of Advanced Fusion Welding Technologies in the Structural Repair of Aluminum and Magnesium Alloys" was approved. This $1.4-million project will evaluate and demonstrate the application of new high-performance welding technologies to the difficult applications of structural weld repair to light alloys in aerostructures and components. Collaborating partners include the University of Manitoba, RRC and Huys Industries.



Engineers in a clean room inspecting a satellite flight panel.

In May, the CARIC regional office held the inaugural CARIC Research Forum for Manitoba and Saskatchewan at the University of Manitoba. Industry, academia and research organizations converged to get an update on the Technology Roadmap, discuss industry's R&D priorities, and consider new research projects.

With the federal government's continued focus on innovation and the National Research Council's new Factory of the Future facility soon to be built in Winnipeg, aerospace innovation in Manitoba has never been stronger. ▮

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